Sepsis Bundle Compliance Tracker: Epic Reporting Translation

> Third featured project, first of the paced SQL series. SQL analysis on a synthetic cohort of 1,800 sepsis encounters across five units, April to September 2026. This document translates the build into Epic reporting terms for a quality committee audience.

Objective

Measure all-or-none compliance with the Surviving Sepsis Campaign 1-hour bundle (SEP-1 style) and identify where the bundle breaks down, by unit, shift, and trend over time, so improvement work can be targeted instead of generic.

Cohort definition

All inpatient and ED encounters with a positive sepsis screen during the measurement period. In Epic terms: patients with a sepsis screening flowsheet row documenting a positive screen, or a Best Practice Advisory (BPA) firing for suspected sepsis, within the encounter date range. Exclusions for a production build would include comfort-measures-only status, sepsis present on arrival at an outside facility with the bundle completed there, and encounters where antibiotics were contraindicated. Those exclusions were not modeled in this synthetic build.

Time zero

Time zero is the moment of sepsis recognition: the timestamp of the positive sepsis screen. Every bundle element is measured forward from this single anchor. In Epic, this maps to the contact timestamp on the screening flowsheet row or the BPA fire timestamp. A single, unambiguous time zero is the most important design decision in the whole report. Everything downstream depends on it.

Bundle elements mapped to Epic data sources

All-or-none: a miss on any single element fails the encounter.

1. **Lactate drawn within 1 hour of time zero.** Source: flowsheet rows for the lactate lab result, using the specimen collection timestamp, not the result timestamp. Collection time is the clinically meaningful moment.

2. **Blood cultures drawn before antibiotics administered.** Sources: blood culture order and collection timestamps compared against the antibiotic administration timestamp. Order time alone is not enough; the comparison must use collection against administration.

3. **Broad-spectrum antibiotics administered within 1 hour of time zero.** Source: MAR administration timestamps. Ordered time and administration time are different events and the bundle measures administration, which is the moment the patient actually receives the drug.

4. **Fluid resuscitation or vasopressors when indicated.** Indicated means hypotension documented after recognition or lactate >= 4.0 mmol/L. Sources: blood pressure flowsheet rows for the hypotension flag, the lactate result for the threshold, intake/output or fluid flowsheet rows for crystalloid volume (>= 30 mL/kg within 1 hour), and MAR administration timestamps for vasopressor initiation.

Translating the logic to Reporting Workbench and SlicerDicer

The SQL build used one base table and a single view (`bundle_compliance`) holding all element logic, with eight analysis queries on top. That maps directly:

- The view becomes the **Reporting Workbench report template**: one row per encounter, with calculated columns for each element flag and the all-or-none result. Element logic lives in one place so a definition change (for example, adopting 3-hour bundle elements) is edited once.
- The unit, shift, and month breakdowns become **report filters and groupings**, or a Radar dashboard with the same slices.

- For ad hoc exploration, **SlicerDicer** can reproduce the cohort (sepsis screen positive encounters) and the time-to-antibiotics distribution, which is useful when the committee asks a question mid-meeting.

Timestamp hygiene and null handling

This is where production builds succeed or fail:

- A missing timestamp means the intervention did not happen. No timestamp, no credit. Nulls are scored as misses, never imputed and never dropped, because dropping them would inflate compliance.
- Administration time versus order time must be kept distinct for antibiotics, cultures, and vasopressors. Mixing them is the most common source of falsely reassuring numbers.
- Lactate must use specimen collection time, not resulted time.
- Fluids require both a start time and a dose (mL/kg); a fluids row with no documented volume cannot satisfy the element.
- About 10% of encounters in real data receive empiric antibiotics before formal recognition. The logic counts those as compliant, which matches clinical intent but should be disclosed in the methodology footnote.

Key findings

- **All-or-none 1-hour bundle compliance: 57.3%** (1,032 of 1,800 encounters).
- **Weakest element: antibiotics within 1 hour at 74.4%.** Cultures before antibiotics (89.6%), lactate within 1 hour (90.7%), and fluids or vasopressors when indicated (93.1%) all perform materially better.
- **ICU leads at 66.5%; Med-Surg trails at 40.1%.** Median time from recognition to antibiotics explains it: 21.7 minutes in the ED versus 46.5 minutes on Med-Surg.
- **Night shift runs 8 points worse than day shift** (53.6% vs 61.3%). The overnight drop is steepest on Med-Surg, falling to 32.7%.
- **Trend is improving:** 53.3% in April to 59.1% in September.
- **Sicker patients get faster care:** compliance is 63.2% for SOFA 9+ versus 54.6% for SOFA 0-4, which raises the question of whether mild presentations are under-triaged.

- **Outcome signal: mortality 9.2% when the bundle was met versus 12.8% when missed.** Unadjusted and synthetic, presented as a signal, not proof.

Limitations

- All data is synthetic. Distributions are realistic but invented, and no real patient data was used.
- The mortality comparison is unadjusted. In production this would require risk adjustment and chart-review validation before being presented as anything stronger than a signal.
- Exclusion criteria (comfort measures, outside transfers, contraindications) were not modeled.
- Single-site style model. A multi-facility build would add site-level variation to the unit breakdowns.

Recommended next steps

1. **Unit-level dashboard** in Radar: bundle compliance by unit and shift, refreshed monthly, with the element-level failure breakdown one click deep. Med-Surg owns the largest opportunity.
2. **Best Practice Advisory for Med-Surg nights:** a real-time alert when a positive sepsis screen has no antibiotic administration documented within 30 minutes, routed to the covering nurse and provider. The data says this is the highest-yield population.
3. **Add 3-hour bundle elements** (repeat lactate, vasopressor application detail) once the 1-hour measure is stable.
4. **Validate the mortality signal** with risk-adjusted analysis and a chart-review sample before citing it beyond the committee.

Synthetic data only. No real patient information was used in this project.